Mixed trace quotient coordinates and a pole-free P-192 chart

Mixed trace -> exact marked translation stabilizer
Add multiplication-map coordinates to generate the quotient field

Quadratic twist shift -> finite rational-input chart
Actual P-192 shift checked; P-192 trace not evaluated

Six targets x eight fixed shifts -> exact finite controls
All collisions retained; common n=6 separates49 conjugates

Derived construction. Let H be the mixed maximal isotropic kernel from round53, of order r^2. Both its projections to E[r] are isomorphisms, so geometrically H is the graph of an anti-isometry psi. Define C(P,R)=sum_(u in E[r]) x(P+u)*x(R+psi(u)). The pointwise evaluator takes the kernel pairs, not the target logarithm. Its trace is invariant under H by reindexing. Because H is Frobenius-stable and x is defined over F_p, the rational function C is defined over F_p even though its summands may involve extension-field torsion. [pcc:unverified]

Exact translation stabilizer. The vertical polar divisors of C are P=-u for all u in E[r]. At such a divisor only the term indexed by u has a pole, and its leading coefficient is x(R+psi(u)). This is a nonzero rational function, so no cancellation removes that divisor. Horizontal divisors similarly identify E[r] in the second factor. If translation by (a,b) fixes C, its polar divisor forces a,b in E[r]. Reindexing u to u+a then compares coefficients x(R+psi(u)+b-psi(a))=x(R+psi(u)). The translation stabilizer of the x-coordinate function on E is trivial, as its unique pole is the identity. Hence b=psi(a). Conversely every (a,psi(a)) fixes C. Therefore the translation stabilizer of C is exactly H. This is a written function-field argument, not a machine-checked universal theorem. [pcc:unverified]

Primitive quotient-field generator. Over a field containing the torsion, let L be the function field of E^2 and N=E[r]^2, acting by translation. The fixed field L^N is generated by the coordinates of [r]P and [r]R. The exact stabilizer makes C have r^2 distinct conjugates under N/H, so L^H=L^N(C). One explicit defining polynomial is product_(v in E[r]) (Z-C(P,R+v)); its coefficients lie in L^N. Thus ([r]P,[r]R,C(P,R)) gives a birational coordinate description of E^2/H on a suitable open chart. This supplies rational quotient coordinates, with degree r^2 over the multiplication-map base. It does not supply a normalized global abelian-surface model or a global group law in these coordinates. [pcc:unverified]

Why the mixed term is needed. The marginal sum over H of x(P+h1) runs once through E[r] and is independent of psi. The same holds for the second factor. Its normalized Velu form is r^2*x([r]P) plus a curve/kernel constant. On the control curve that constant is zero, and the identity is checked on all nine shifted arguments. The mixed trace retains the coupling that these marginal traces lose. [pcc:unverified]

Pole-free rational-input chart. The unshifted trace has polar divisors at r-torsion in each coordinate. Choose S,R0 over F_(p^2) with [r]S,[r]R0 nonzero and evaluate C(P+S,R+R0). If P,R are rational order-r points, every summand is finite: a vanishing translated argument would imply S or R0 belongs to E[r]. Fixed translation preserves the kernel stabilizer and the quotient-field description. Since the shifted input lies over F_(p^2) and the trace is defined over F_p, its value is in F_(p^2). [pcc:unverified]

Actual P-192 shift. With W^2=-1, p=3 mod4 makes W nonrational and W^p=-W. The first small x with nonsquare curve RHS is x=1. A square root y0 of minus that RHS gives S=(1,y0*W), a nonzero point with pi(S)=-S. The producer verifies the quadratic-field curve equation, twist-order annihilation, and nonzero [r]S and [r](6S). A separate checker performs the scalar calculations on the base-field quadratic twist, using (X,Y)->(-X,YW), and matches all recorded points. The twist order is coprime to r and to six. Hence (S,6S) supplies an actual pole-free shift for all rational P-192 input pairs. The P-192 trace itself has not been evaluated. [pcc:unverified]

Exact small-field evaluator. The six kernels of round53 are embedded in F_(11^6)=F_(11^3)[W]/(W^2-2). On E/F_11:y^2=x^3+2x+7, S=(0,3W) has order17 and pi(S)=-S. The frozen panel evaluates (S,nS) for every n=1,...,8. For each target it computes all49 conjugate traces, their rational-coset subset, and direct H-translation controls. All outcomes are retained, including collisions. The n=1 choice has only27 or28 conjugates for two targets. The common n=6 specialization, selected after the complete panel, has49 distinct conjugates and seven distinct rational cosets for every target; this is an exact control choice, with no statistical inference. [pcc:unverified]

Distinct conjugates: six target kernels, all eight fixed shifts

	n=1	n=2	n=3	n=4	n=5	n=6	n=7	n=8
Target 1	27	49	49	49	47	49	49	46
Target 2	28	49	47	49	47	49	49	49
Target 3	49	48	49	49	49	49	49	49
Target 4	49	47	49	48	49	49	49	49
Target 5	49	49	46	49	49	49	49	49
Target 6	49	47	49	49	48	49	48	48

Exact finite-field counts; all outcomes retained. n=6 separates all49 in every row.

Norm and Galois checks. Both the initial specialization, including repeated roots, and the separating n=6 specialization retain their full monic degree49 norm polynomials. Their coefficients are in F_11. For n=6 the distinct root orbits have degrees 1,2,2,2,6,6,6,6,6,6 under Frobenius, accounting for all49 roots. Rational-coset trace values are in F_(11^2), with Frobenius sending index i to -i. Across the fixed panel the49 conjugate signatures are distinct, and the joint translation stabilizer among all2401 r-torsion translations is exactly the49-point kernel. Finite specialization collisions are preserved rather than interpreted as a failure of the generic argument. [pcc:unverified]

Independent replay. The separate checker uses packed cubic-field elements, independent polynomial reduction and a separate quadratic-tower and curve implementation. It recomputes all2352 trace values, nine marginal identities, all twelve degree49 norm polynomials and the common separating fibers. The P-192 shift has an independent base-field twist verifier. Kernel points originate from public-point construction in round53; no target logarithm is read by the trace producer. [pcc:unverified]

Cost and remaining interface. With translated x-coordinate arrays available, one raw trace uses r^2 products and r^2-1 additions in the working field. Computing a complete conjugate family or an expanded norm introduces further work. No P-192 speed claim follows from the quotient-field generator. The next algorithmic target is compressed trace/norm evaluation or normalization that uses the CM structure without materializing the r^2-point algebra. The present trace evaluator and pole-free chart make that interface concrete. [pcc:unverified]

Literature audit. Lubicz and Robert, Computing isogenies between abelian varieties, Algorithm6.1 and analysis6.2 (https://arxiv.org/pdf/1001.2016), take a kernel basis in theta coordinates. They reconstruct compressed theta null data using logarithmic-length addition chains and root extractions; full decompression has a degree-dependent cost. The full kernel basis and its field/theta representation are input obligations. Our compact kernel equations are not yet that input basis. This identifies a second concrete route to investigate, rather than treating the r^2-term trace as a bound on all quotient algorithms. Invariant averaging and Galois fixed-field generation are standard; no literature novelty claim is made for them.

Next construction. Develop either a compact mixed-trace evaluator retaining the marking, or an adapter from the Miller/Frobenius kernel presentation to the compressed-theta input. Any cost must include kernel realization, theta lifts/root extraction or trace algebra, output normalization and the scalar-extraction interface.